M|P|W

Physics A

Chapter 4
The Physics Department

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **34**.
- The total number of marks may take into account some 'either/or' question choices.

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

1. The table below shows the measurements recorded by a student for a solid metal sphere. The absolute uncertainties in the mass of the sphere and in its radius are also shown.

mass	100 ± 6 g
radius	1.60 ± 0.08 cm

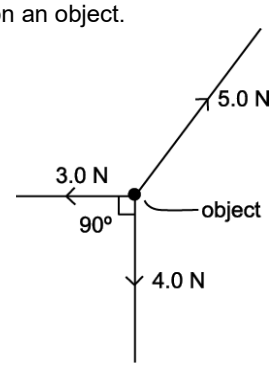
What is the percentage uncertainty in the density of the sphere?

A. 1%
B. 11%
C. 16%
D. 21%

Your answer ☐

[1]

2. The diagram below shows three forces acting on an object.



The object is **stationary.** All the forces lie in the vertical plane. The weight of the object is 4.0 N.

Which statement is **not** correct?

A. The resultant force on the object is zero.
B. The magnitude of the resultant force of 3.0 N and 4.0 N forces is 5.0 N.
C. The magnitude of the vertical component of the 5.0 N force is 4.0 N.
D. The resultant force in the horizontal direction is 3.0 N.

Your answer ☐

[1]

3. Which physical quantity has the same base units as energy?

A. moment
B. momentum
C. force
D. pressure

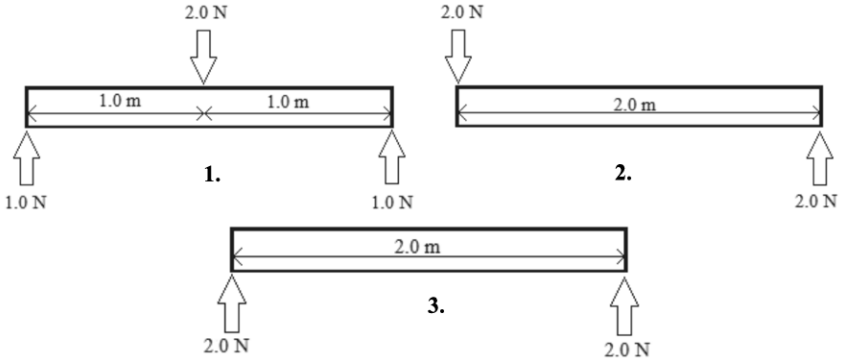
Your answer ☐

[1]

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

4. A 2.0 m rigid rod with negligible weight is subject to forces in three different ways as shown in diagrams **1-3** below.



For the rod to be in equilibrium which of the diagrams above is / are correct?

A. 1, 2 and 3
B. Only 1 and 2
C. Only 2 and 3
D. Only 1

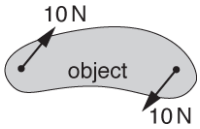
Your answer ☐

[1]

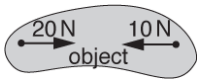
5. Two forces act on an object in the same plane.

Which diagram shows a couple?

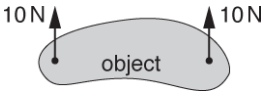
A



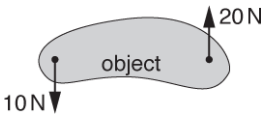
B



C



D



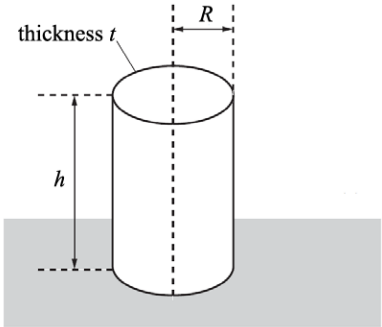
Your answer ☐

[1]

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

6. A group of civil engineers are assessing whether or not to use solid concrete pillars or hollow metal tubes to support a building. One such tube is shown below. The tube is placed on a horizontal surface. The tube is made of metal of thickness *t*. The tube has height *h* and a mean internal radius *R*. The radius *R* >> thickness *t*.



A heavy metal block of mass *m* is placed on top of the tube.

What is the approximate pressure *p* acting on the tube?

A $p = \frac{mg}{2\pi Rt}$

B $p = \frac{mg}{\pi R^2}$

C $p = \frac{mg}{\pi R^2 h}$

D $p = \frac{mg}{\pi R^2 t}$

Your answer ☐

[1]

7. A brick of mass *m* has sides of lengths *a, b* and *c,* where *a* < *b* < *c*. The brick is placed on a horizontal table such that the pressure it exerts on the table is a maximum.

What is the maximum pressure *p* acting on the table?

A $p = \frac{mg}{ab}$

B $p = \frac{mg}{ac}$

C $p = \frac{mg}{bc}$

D $p = \frac{mg}{abc}$

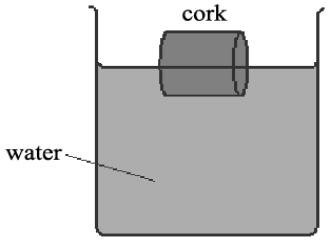
Your answer ☐

[1]

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

8. A bottle cork floats on water. It is partially submerged in the water.



Which of the following statements is / are true?

1. The net force acting on the cork is equal to the weight of the water displaced.
2. The weight of the cork is equal to the upthrust on the cork.
3. The upthrust on the cork is equal to the mass of the water displaced.

A. 1, 2 and 3
B. Only 2 and 3
C. Only 3
D. Only 2

Your answer ☐

[1]

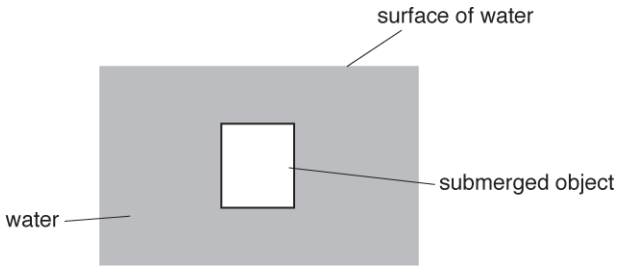
9. A small block of wood is placed in deep water.
The block is at rest with 80% of its volume under the surface of water.
The weight of the block is 6.0 N.

What is the upthrust acting on the block?

A 0
B 1.2 N
C 4.8 N
D 6.0 N

Your answer ☐ **[1]**

10. The diagram below shows an object submerged in water.



© OCR 2023. You may photocopy this page.

Created in ExamBuilder

The object is stationary in the water.

Which statement about the upthrust acting on the object is correct?

A It is zero.

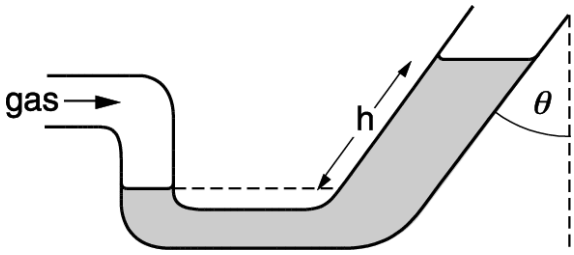
B It is equal to the mass of the object.

C It is equal to the weight of the object.

D It is equal to the volume of the water displaced.

Your answer ☐ **[1]**

11. Gas under pressure is forced into a pipe formed into a U shape bend. A liquid of density ρ is pushed up the right hand side so there is a difference h in the height on each side, measured along the side of the tube.
The pipe on the right hand side is inclined at an angle θ to the vertical and it has a cross sectional area double that on the left side.



What is the pressure of the gas in excess of atmospheric pressure?

A. $h\rho g$
B. $2h\rho g$
C. $h\rho g \cos\theta$
D. $2h\rho g \cos\theta$

Your answer ☐

[1]

12. What is the best estimate for the density of steel?

A. 10^{-3} g cm^{-3}
B. 10^{3} g cm^{-3}
C. 1 kg m^{-3}
D. 10^{3} kg m^{-3}

Your answer ☐

[1]

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

13. i. Define the moment of a force about a point.

[1]

ii. State the principle of moments as a condition for equilibrium.

[1]

14. A block of concrete rests on the ground, as shown in Fig. 7.1.

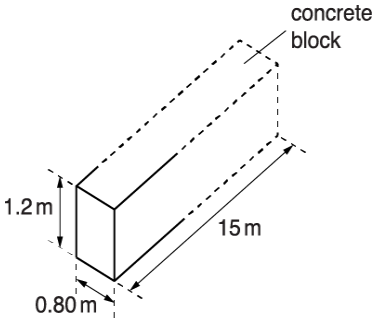


Fig. 7.1

The concrete block is 15 m long, 0.80 m wide and 1.2 m high. The density of concrete is 2.4×10^3 kg m^{-3}. Calculate

i. the weight of the concrete

weight = ...

ii. the pressure exerted on the ground by the block of concrete.

pressure = ...

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

15. Sirius A and B are binary stars in our galaxy at a distance of 8.6 ly from the Sun. Sirius B is a white dwarf of diameter 12 km and mass 2.0×10^{30} kg.

Calculate the density of Sirius B.

density = ... unit ..... **[3]**

16. Fig. 2.1 shows an object held horizontally by a string.

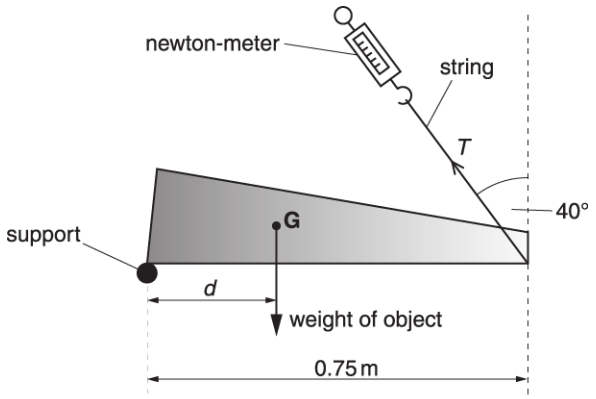


Fig. 2.1

One end of the object rests on a support and the other end is held by the string. The object is in equilibrium. Point **G** is the centre of gravity of the object. The mass of the object is 1.2 kg. The tension *T* in the string is 5.1 N. The string makes an angle of 40° with the vertical.

Take moments about the support and calculate the distance *d*.

d = ... m **[3]**

17. Fig. 22 shows a uniform platform secured to a wall and resting on a vertical concrete pillar.

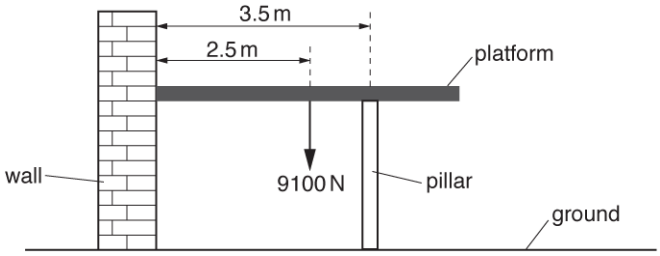


Fig. 22

The platform is in a horizontal position.

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

The weight of the platform is 9100 N and it has length 5.0 m. The centre of the pillar is 3.5 m from the wall.

Use the principle of moments and the information provided in Fig. 22 to calculate the vertical force *F* exerted by the pillar on the platform.

F = ... N **[2]**

18(a). Fig. 23.1 shows a metal cylinder of diameter of about 5 cm placed on a horizontal table.

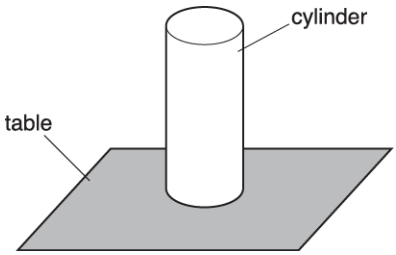


Fig. 23.1

i. State *Archimedes' principle.*

[1]

ii. Fig. 23.2 shows the metal cylinder hung from a newtonmeter.

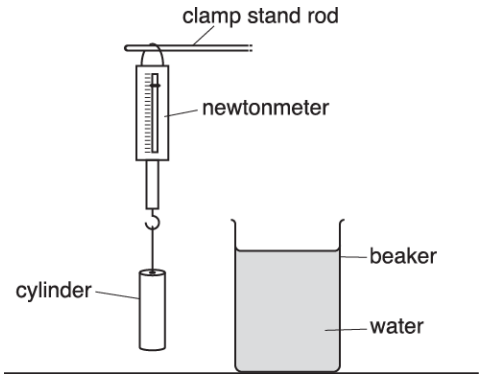


Fig. 23.2

The reading on the newtonmeter is 9.0 N.
The cylinder is slowly lowered into water in a beaker until it is completely submerged.
The cylinder does not touch the side or the bottom of the beaker. The newtonmeter reading now is 7.8 N. The density of water is 1000 kg m^{-3}.
Calculate the density ρ of the metal of the cylinder.

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

ρ = ... kg m^{-3} **[3]**

(b). Fig. 23.1 shows a metal cylinder of diameter of about 5 cm placed on a horizontal table.

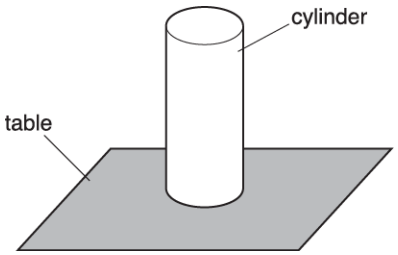


Fig. 23.1

Describe how you can use instruments available in a physics laboratory to determine the pressure exerted by the cylinder on the table. State how you would make your results as precise as possible.

[4]

END OF QUESTION PAPER

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

Mark scheme

Question			Answer/Indicative content	Marks	Guidance
1			D	1	
			Total	**1**	
2			D	1	
			Total	**1**	
3			A	1	
			Total	**1**	
4			D	1	
			Total	**1**	
5			**A**	**1**	
			Total	**1**	
6			A	1	
			Total	**1**	
7			A	1	
			Total	**1**	
8			D	1	
			Total	**1**	
9			**D**	1	
			Total	**1**	
10			**C**	**1**	
			Total	**1**	
11			C	1	
			Total	**1**	
12			D	1	
			Total	**1**	
13		i	Force × perpendicular distance from pivot / fulcrum	B1	

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

		ii	Clockwise moments = anticlockwise moments about any axis **or** zero resultant moment about any axis	B1	**Allow** alternatives such as about a given point
			Total	**2**	
14		i	mass = 2400 × (0.80 × 1.2 × 15) / mass = 3.46 × 10^4 (kg)	C1	
		i	weight = 3.46 × 10^4 × 9.81		
		i	weight = 3.4 × 10^5 (N)	A1	**Examiner's Comments** Generally this was well answered with most candidates correctly using the equations for density and weight. Some candidates stopped after calculating the mass of the concrete block, perhaps thinking that this was the '*weight*' of the block.
		ii	pressure = 3.4 × 10^5/(15 × 0.80)	C1	Possible ecf from **(a)(i)**
		ii	pressure = 2.8 × 10^4 (Pa)	A1	**Examiner's Comments** Most candidates gained two marks for determining the pressure. Only a very small number of candidates used the incorrect value for the cross-sectional area.
			Total	**4**	
15			$V = \frac{4}{3}\pi \times (6\times10^3)^3$ or V = 9.05 × 10^{11} (m^3)	C1	**Note** : An incorrect equation here for V prevents this and any subsequent marks.
			density = $\frac{2.0\times10^{30}}{\frac{4}{3}\pi\times(6\times10^3)^3}$	C1	
			density = 2.2 × 10^{18} kg m^{-3}	A1	The correct unit must also be included to score this A1 mark. **Allow** 2 marks for 2.76.... × 10^{17} kg m^{-3} - 12 km used instead of 6 km for the radius. **Examiner's Comments** The vast majority of the candidates correctly calculated the density of Sirius B to be 2.2 × 10^{18} kg m^{-3}. A very small number of candidates incorrectly calculated the volume of the star using the equation V = $4\pi r^2$.
			Total	**3**	

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

16			5.1 × cos40 × 0.75 or d × 1.2 × g	C1	
			5.1 × cos40 × 0.75 = d × 1.2 × 9.81	C1	
			$d = \frac{5.1 \times \cos 40 \times 0.75}{1.2 \times 9.81}$		
			d = 0.25(m)	A1	**Allow** 2 marks if sine of the angle is used instead of cosine; this gives an answer of 0.21 (m) **Allow** use of 9.8 (m s^{-2}) **Note**: '5.1 × 0.75 = d × 1.2 × 9.81; d = 0.32… (m)' 1 mark because of the first C1 **Examiner's Comments** Candidates, across the ability spectrum, did well with this question. Candidates demonstrated an excellent knowledge of the principle of moments and the resolving a vector (the tension T in the string). The most common errors were using sin40° instead of cos40° and not converting the mass of 1.2 kg into the weight of the object when taking moments about the support.
			Total	**3**	
17			(clockwise moment = anticlockwise moment) 2.5 × 9100 = 3.5 × F (Any subject) F = 6500 (N)	**C1** **A1**	**Examiner's Comments** Most candidates effortlessly applied the principle of moments to calculate the vertical force at the pillar. A few candidates took moments about the pillar, determined the force on the platform at the wall and then calculated the force F at the pillar using 'net vertical force = 0'. Although this was a longer route, it was still worthy of two marks. About a quarter of the candidates scored nothing. The common errors were quoting the moment of the weight about the wall (22750 Nm) as the force F and using 5.0 m instead of 3.5 m in the calculations.
			Total	**2**	
18	a	i	The upthrust is equal to the weight of the fluid / liquid / water / air displaced	B1	**Examiner's Comments**

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

					About one in every seven candidates omitted this question and only about a third of the candidates gave an acceptable statement of Archimedes' principle. It was clear from the answers that most candidates had not revised this topic. There were countless guesses, with many famous laws incorrectly linked to this principle.
		ii	(upthrust =) 9.0 − 7.8 (N) or (mass =) 9.0/9.8(1)	C1	**Note**: This C1 mark for determining the upthrust (1.2 N) or the mass (0.92 kg) of the cylinder
		ii	$V = \frac{(1.2/9.81)}{1000}$ or $V = 1.2(23) \times 10^{-4}$ (m^3) $\rho = \frac{(9.0/9.81)}{1.223 \times 10^{-4}}$	C1	
		ii	$\rho = 7.5 \times 10^3$ (kg m^{-3})	A1	**Allow** full credit for alternative methods, e.g: $\rho = \left(\frac{9.0}{1.2}\right) \times 1000 = 7.5 \times 10^3$ (kg m^{-3}) **Examiner's Comments** This proved to be a discriminating question that favoured those candidates who could apply, rather than just rote learn, Archimedes' principle. About a third of the candidates scored nothing in this question but many candidates did score one mark for determining the upthrust of 1.2 N. Most candidates stopped at this point. The top-end candidates correctly determined the volume of the displaced water and then went on to successfully calculate the density of the metal.
	b				**Note**: In this question any symbols used must be defined or previously mentioned **Note**: Allow full credit for alternative methods, e.g. using the equation pressure = height × density × *g*
			pressure = $\frac{\text{weight (of cylinder)}}{\text{area}}$	B1	**Allow** force / area
			Weight (of cylinder) determined using a newtonmeter or Measure mass (of cylinder) using balance / scale(s) and multiplying by *g* / 9.8(1 m s^{-2})	B1	**Not** 'gravity' for *g*
			Area determined by measuring the diameter with a ruler / vernier callipers / micrometer and then using (area =) π × r^2	B1	**Not** measure radius **Allow** other correct methods

© OCR 2023. You may photocopy this page.

Created in ExamBuilder

			A sensible suggestion that reduces the % uncertainty: Use micrometer / (vernier) calipers / travelling microscope Use balance / newtonmeter with smaller division (AW)	B1	**Not** 'repeat readings (of diameter etc.)' because this procedure improves the accuracy and not the precision **Allow** balance / newtonmeter with 'high resolution' **Examiner's Comments** The majority of the candidates scored two or more marks for this practical based question. It is good to report that many candidates were familiar with instruments used to measure diameter and mass. Vernier calipers and micrometers were mentioned by some candidates for measuring the diameter of the metal cylinder. Instead of using the equation pressure = force / area, a small number of candidates successfully gained some marks by applying the equation $p = h\rho g$. A small number of candidates confused the terms *weight* and *mass*. Only a very small number of candidates realised that precision was linked to reducing the percentage uncertainty in the final value of the pressure. Answers such as '*Use a micrometer measuring to ± 0.01 mm instead of a ruler marked in mm*' or '*Use a digital balance giving mass to the nearest 0.01 g instead of 1 g*' were awarded the final mark. Most candidates however, confused accuracy with precision and went on to describe how the experiment could be made precise by taking multiple readings of diameter or mass. A significant number of candidates omitted answering the precision part of the question. It is worth reminding candidates that it is important to carefully examine the question before writing their answers.
			Total	**8**	

© OCR 2023. You may photocopy this page.

Created in ExamBuilder